A Neuro-Symbolic Expert System for Medical Diagnosis

From Natural Language to Explainable Logic — Rule-Based Inference for Endemic Febrile Illness Differentiation in Low-Resource Clinics

NEURO-SYMBOLIC AI | EXPERT SYSTEMS | CLINICAL DECISION SUPPORT | EXPLAINABLE AI | GLOBAL HEALTH TECH

Bridging the Gap Between Neural Perception and Symbolic Reasoning in Clinical AI

The Problem

The Black Box Crisis

Deep models achieve high accuracy but are opaque; clinicians cannot trace diagnoses, making them unsafe for high-stakes decisions.

The Rigidity of Rule-Based Systems

Symbolic systems are transparent but brittle: they demand rigid inputs and fail on unstructured, conversational data.

The Neuro-Symbolic Solution

A three-tier architecture decouples natural language perception from deterministic medical logic; the LLM only extracts information while a validated JSON KB handles reasoning.

The system accepts natural language, validates extracted facts, and emits auditable, explainable diagnoses.

Natural Language → **Neural Extraction** → **Symbolic Inference** → **Explainable Output**

A Three-Tier Architecture That Guarantees Explainability at Every Step

Tier 1: Neural Layer

The LLM parses unstructured patient text and outputs a structured JSON dictionary of recognized symptoms with semantic Certainty Factors. Constrained prompting ensures it acts solely as a data extractor.

Tier 2: Integration Layer

Raw JSON facts are validated against a hardcoded medical vocabulary. Unknown symptoms are dropped to eliminate hallucinations. Conversational phrases map to strict ontological variables.

Tier 3: Symbolic Layer

A Python Evidence-Weighted DAG engine applies MYCIN-style probabilistic mathematics to calculate diagnostic certainty dynamically.

```
AND: CF_rule =
min(CF_s1...CF_sn) ×
Rule_Base_CF
OR: CF_combine = CF_old + CF_new
× (1 − CF_old)
```

Knowledge Base Extract (Evidence-Based Mapping)

Disease	Pathognomonic Symptoms	Base CF
Malaria	Cyclical fever (48h), Severe chills, Heavy sweating, Jaundice	0.90
Typhoid	Gradually increasing fever, Rose spots rash, Abdominal pain	0.85
Dengue	Sudden high fever (40°C), Retro-orbital pain, Severe joint pain	0.90
Lassa Fever	Mucosal bleeding, Facial swelling, Sudden deafness	0.95

Full Epistemological Transparency — Every Diagnosis Comes With a Traceable Audit Trail

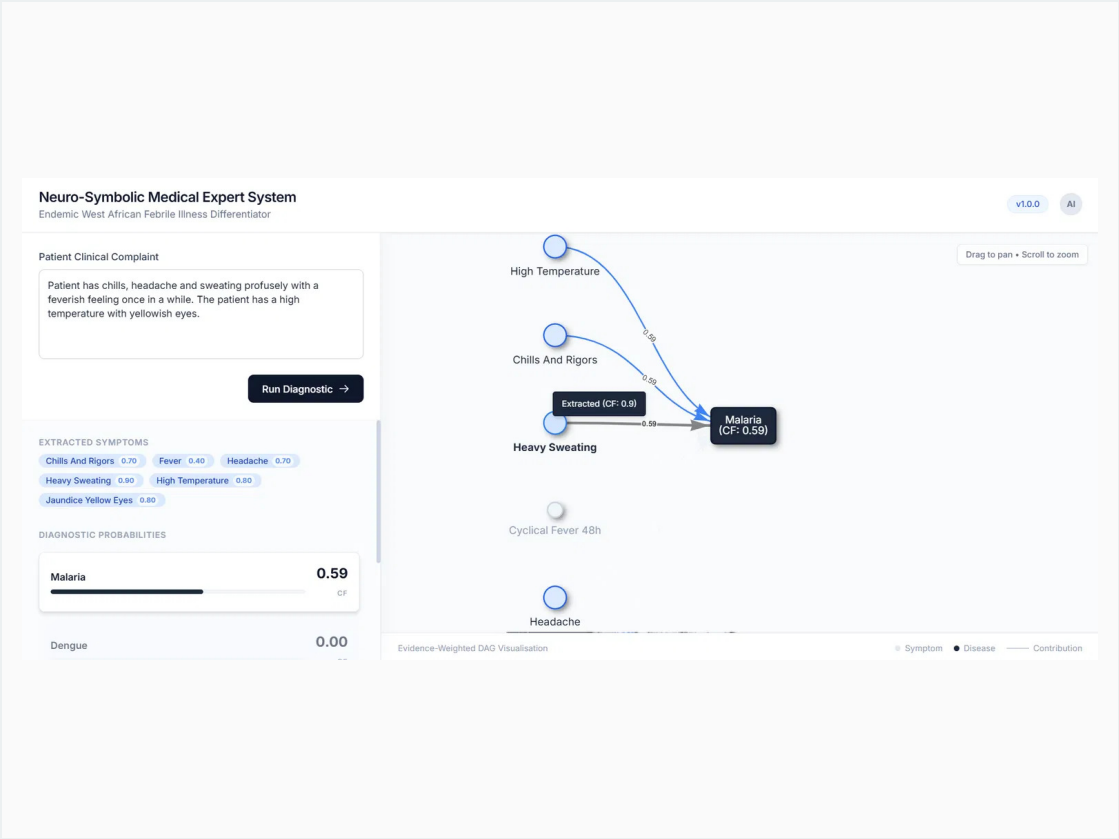

Fig 1. The Evidence-Weighted DAG Visualisation Interface

Explainability by Design

The system builds a real-time JSON audit trail as the DAG inference engine runs. Rendered as an interactive visual graph, symptoms appear as root nodes, rules as weighted edges, and diseases as sink nodes — giving clinicians a complete, traceable reasoning path.

Why This Matters for Global Health

Engineered specifically for under-resourced clinics in West Africa, where laboratory infrastructure is limited. Accurate differential diagnosis of overlapping febrile illnesses (Malaria, Typhoid, Dengue, Lassa Fever) can be life-saving using only natural language symptom descriptions.

Core Achievement

By strictly confining deep neural networks to feature extraction and utilising an Evidence-Weighted DAG for mathematical inference, the system completely prevents AI hallucinations while providing highly competitive diagnostic accuracy and clinical transparency.